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Direct femtosecond laser welding of silica glass for perovskite film encapsulation

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Highlights

- This study pioneers a novel encapsulation strategy for perovskite films by demonstrating, for the first time, direct femtosecond laser welding of silica glass to ITO glass.
- By controlling the laser parameters, a micron-scale melting-recrystallization process was induced at the interface, achieving a shear strength of up to 147.6 MPa.
- The encapsulated perovskite films showed no signs of degradation after thermal cycling (from 85°C to -20°C) and 168 hours of water immersion.

- Femtosecond laser encapsulation provides a new approach for enhancing the long-term stability of optoelectronic devices, particularly perovskite solar cells.

Abstract

The environmental sensitivity of perovskite solar cells (PSCs) significantly hinders their commercial viability. A currently employed method for encapsulating perovskite solar cells is the use of potting compounds. However, adhesive-based encapsulation suffers from inherent drawbacks, such as susceptibility to aging and volatilization, which ultimately compromise the long-term reliability, stability, and lifetime of the solar cells. This study developed a novel method using a femtosecond laser (1030nm wavelength, 300fs pulse width) to directly weld silica glass for perovskite film encapsulation. By optimizing processing parameters (single-pulse energy: 1.37 μ J, repetition rate:1 MHz), localized melting and recrystallization were induced at the indium tin oxide (ITO)/silica glass interface, achieving covalent bonding. The optimized weld lines exhibited a high shear strength of 147.6MPa and formed a continuous molten transition zone at the interface. This high strength indicates excellent mechanical stability of the laser-welded encapsulation. After enduring 10 thermal cycles (between 85°C and -20°C) and 168h of water immersion, the encapsulated samples maintained structural integrity without color degradation or decomposition. These results highlight the reliability of this encapsulation method under extreme environmental conditions. To our knowledge, this work represents the first demonstration of long-term hermetic sealing of perovskite films using femtosecond lasers, overcoming the limitations of traditional adhesive encapsulation (prone to aging and volatilization) and the insufficient stability (less than 6h) of picosecond laser sealing. This work not only provides a highly reliable encapsulation solution for environmentally sensitive optoelectronic devices, but also paves the way for the monolithic integration and commercialization of perovskite-based optoelectronics.

CRediT authorship contribution statement

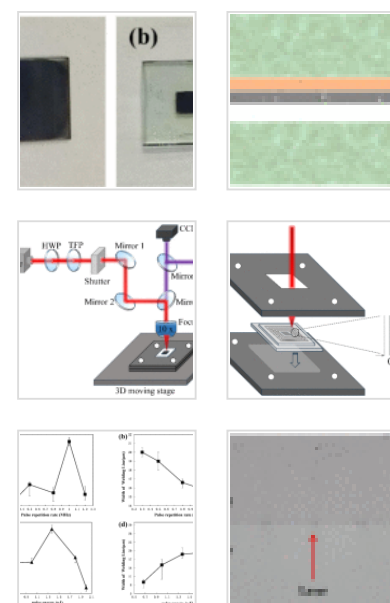
Declaration of competing interest

Acknowledgments

Data availability

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Keywords

Femtosecond laser; Perovskite cell; Silicaglass encapsulation; Laser welding; Interface bonding

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1. Introduction

Perovskite solar cell is considered promising candidates for next-generation photovoltaic technologies because of their high power conversion efficiencies (PCEs), reaching 27.3% for single-junction and 33.9% for tandem configurations, and their potential for low manufacturing costs [1], [2], [3], [4], [5], [6]. However, their commercial viability is severely impeded by stability issues due to their sensitivity to environmental factors such as humidity, oxygen and light [7], [8]. For example, methylammonium lead iodide (MAPbI_3) undergoes hydrolysis-induced yellowing at relative humidity (RH) levels exceeding 50% [9]. Consequently, the development of robust, long-term encapsulation technologies is of paramount importance [10], [11]. Traditional encapsulation strategies exhibit limitations: (i) Adhesive encapsulation uses polymers such as epoxy resin [12], butyl rubber [13], or polydimethylsiloxane (PDMS) [14] to bond and seal the cover glass to the solar cell substrate. However, these polymers are prone to aging and volatilization, which can lead to seal failure. (ii) Film barrier layers, formed via atomic layer deposition (ALD), create nanoscale metal oxide barriers (e.g., Al_2O_3 [15], SnO_x [16], SiO_x [17]) or use organic composite films, such as the flexible organic barrier layer [18]. Nevertheless, the fabrication processes for both types of barrier layers are relatively complex and costly. Despite their ability to encapsulate, current sealing methods lack the long-term durability necessary to achieve the commercially mandated 20-year service life.

Although they offer novel approaches for PSC encapsulation, emerging ultrafast laser technologies are hindered by critical bottlenecks: While Ribeiro et al. (2012) pioneered laser-assisted frit sealing for dye-sensitized solar cells (DSCs) [19], the method necessitates pre-deposited intermediate layers. Jorge et al. (2020) applied laser-assisted glass frit encapsulation to PSCs without a hole transport layer (HTM) [20]. Nevertheless, their dual-laser beam process (combining preheating and localized melting) can result in issues such as incomplete melting and localized overheating, potentially causing thermal degradation of the perovskite [21]. Under picosecond laser processing, Chen et al. (2021) conducted sealing experiments on glass substrates coated with perovskite films using quartz glass and soda-lime glass, exhibiting only 2-hour water resistance with quartz glass and 6-hour resistance with soda-lime glass [22], long-term stability remains a concern. In related welding processes, the excessive heat-affected zone generated by picosecond laser welding led to increased thermal stress, which adversely affected the welding quality and compromised sealing performance.

Nanosecond laser processing operates through linear absorption and thermal diffusion, leading to a significant heat-affected zone (HAZ) that often induces microcracks and residual stress in glass [23]–[24]. In contrast, femtosecond lasers welding, leveraging its ultra-short pulse characteristics and ultra-high peak power (approximately 10^{14} W/cm²), enable precise control over micrometer-scale (below 100 μm) melting processes [25]–[26]. This significantly reduces the heat-affected zone, mitigates thermal stress, and lowers the risk of damage to thermosensitive materials like perovskite, thereby enhancing long-term weld stability. Moreover, utilizing long-focal-depth, diffraction-free beams under non-tight focusing conditions can induce near-field nanoscale material ablation [27]. Furthermore, the nonlinear absorption induced by femtosecond pulses facilitates glass micro-zone melting and bonding, preventing thermal damage and offering benefits such as eliminating intermediate layers, ensuring high precision, and achieving strong bonding strength [28], [29], [30].

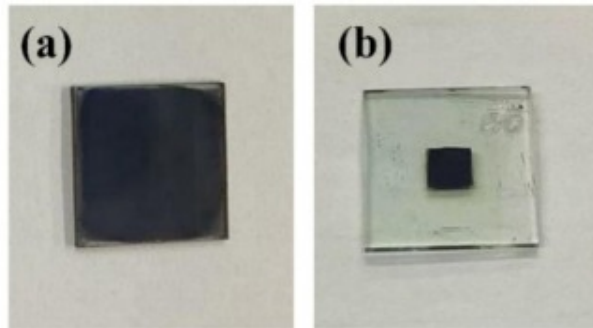
To the best of our knowledge, we report the first direct femtosecond laser welding of silica glass to ITO glass for encapsulating perovskite films, effectively overcoming the

limitations of picosecond laser-based techniques. Through optimized laser parameters (1.37 μJ , 1 MHz), we achieved micron-scale interfacial melting-recrystallization and covalent bonding, resulting in a record shear strength of 147.6 MPa and exceptional hermetic performance—showing no degradation after 168 h of water immersion and 10 thermal cycles (85°C to −20°C). This robust approach overcomes the limitations of conventional adhesive and picosecond-laser-based methods, providing a reliable encapsulation solution for highly sensitive optoelectronic devices.

2. Materials and experimental setup

The perovskite active layer, which forms the core light-harvesting component of PSCs, is highly sensitive to environmental temperature and humidity. In this experiment, perovskite films were used instead of full PSC devices for direct femtosecond laser welding. The key mechanism of femtosecond laser welding lies in nonlinear absorption-induced localized micro-melting, a process universally applicable to various transparent dielectric materials, including multiple glass types [31]–[32]. This technique has been successfully demonstrated across multiple material systems, including: fused silica-borosilicate joining by Watanabe et al. [29], fused silica-BK7 welding achieving 5.25 MPa strength by H  lie et al. [33], and subsequent extensions to metal-glass and silicon-glass combinations [34], [35], [36], [37]. While this study utilizes silica glass as a model substrate due to its optimal transparency and thermal stability, the methodology can be extended to other glass types including borosilicate and aluminosilicate glasses. The substrate used in this experiment was ITO-coated silica glass (15 mm $\times$ 15 mm $\times$ 1 mm; ITO thickness: 500 nm). Cover plates of the same dimensions were made of silica glass. A 1.4 M $\text{Cs}_{0.05}\text{FA}_{0.95}\text{PbI}_3$ perovskite solution, prepared in the laboratory, was spin coated on the ITO substrate and subsequently annealed at 100°C for 10 min (Fig. 1(a)), resulting in a film approximately 300 nm thick. For encapsulation, the peripheral perovskite was removed with ethanol, leaving a central 6 mm $\times$ 6 mm area (Fig. 1(b)). The schematic diagram of the sample sandwich structure is shown in Fig. 2. The sample features a typical sandwich-like configuration, with the perovskite layer serving as the functional core, encapsulated

between an ITO transparent conductive layer and a glass substrate. The laser was selectively applied to the peripheral region where the perovskite film had been removed (Fig. 1(b)), thereby creating a protected sealed cavity for the central perovskite layer.



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Fig. 1. Perovskite film spin-coated on ITO glass substrate. (a) Original sample, (b) after erasing the peripheral perovskite material.



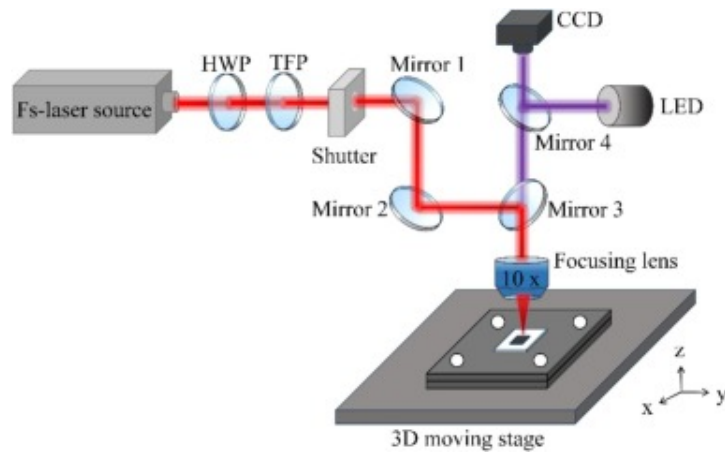
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Fig. 2. Schematic diagram of sample sandwich structure.

The experimental setup for femtosecond laser welding is illustrated in Fig. 3. An amplified Ti:sapphire laser system emitting 1030 nm, 300 fs, Gaussian laser pulses, was used in the experiment. The pulse repetition rate is tunable between 1 Hz and 1 MHz. The half wave plate (HWP) combined with a thin film polarizer (TFP) is employed to adjust the pulse energy. A mechanical shutter, synchronized with the laser system, enabled accurate

control over the laser transmission and blocking timing. A CCD camera is placed along the optical axis and used to align the sample and obtain a live view for the laser processing. After reflection by the mirror, the laser beam is focused onto the sample mounted on the fixture by a 10× objective lens. The measured laser spot diameter was 5.1 μm . The motion system includes an automated high-accuracy three-dimensional (x, y, z axis) linear stage with 150mm travel range and 10mm/s maximum speed.



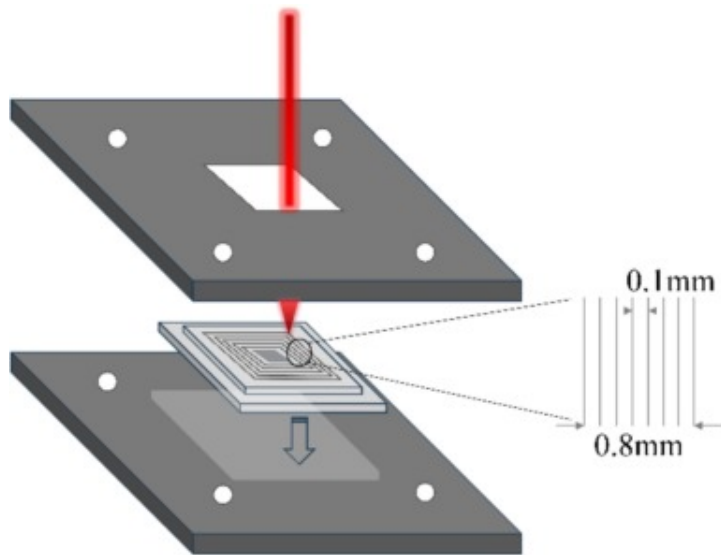
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Fig. 3. Femtosecond laser experimental setup diagram.

To ensure effective femtosecond laser welding at the quartz/ITO glass interface, intimate contact between the mating surfaces is essential. To achieve this, a custom-designed precision clamp (Fig. 4) was used to securely hold the samples and apply the required contact pressure [30]. During the welding process, the femtosecond laser beam was directed vertically through an optical access window at the top of the clamp, targeting the sample interface. The welding path was configured as a square closed contour with a line spacing of 0.1 mm, employing a layer-by-layer scanning strategy from the outer region inward, where the outermost weld line was completed first, followed by rapid cooling and solidification. This approach yielded three significant advantages: (i) the perimeter

welding strategy provided spatial isolation from functional layers, while the femtosecond laser's minimal heat-affected zone and non-thermal processing mechanism prevented damage to adjacent organic materials; (ii) it facilitated the closure of initial gaps at the interface; and (iii) it created a rigid “skeleton” structure through the pre-solidified outer weld seam. As the laser scanned the inner regions, the constraint effect provided by this rigid skeleton effectively mitigated the thermally induced stresses arising during the subsequent welding process, thereby substantially reducing thermal deformation in the sample.



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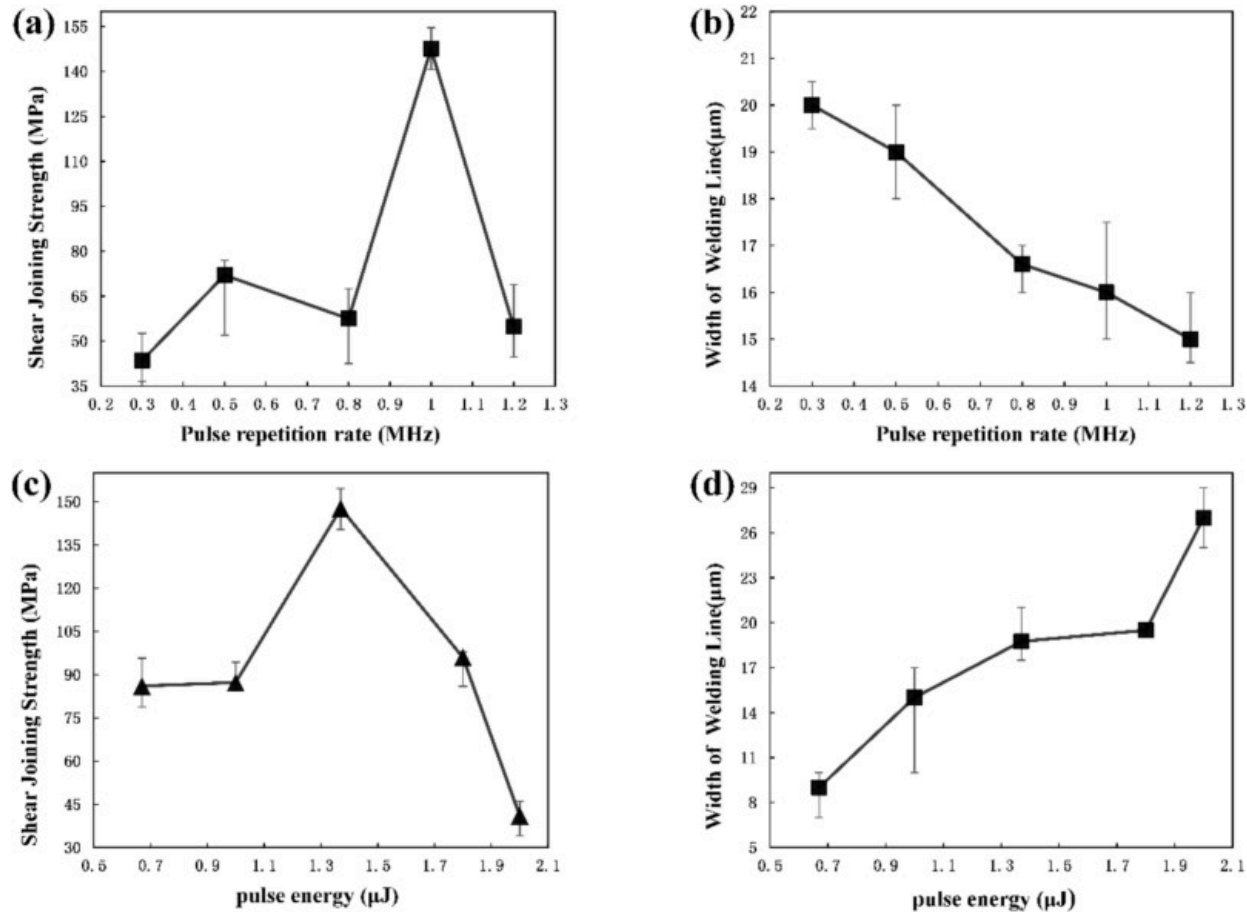
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Fig. 4. Schematic of the precision positioning assembly for silica/ITO glass laser welding.

3. Experimental results

The femtosecond laser was operated with a pulse width of 300fs, a scan speed of 2 mm/s, and a line spacing of 0.1 mm. Under these parameters, the effects of laser pulse energy and repetition rate on welding quality were investigated, and the corresponding results

are presented in Fig. 5. Note that the values of each point in the curves take an average of three measurement results to reduce the errors caused in the welding processes and measurements.



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Fig. 5. (a) Shear strength and (b) weld line width as a function of pulse repetition rate; (c) shear strength and (d) weld line width as a function of pulse energy.

Fig. 5 (a) and (b) illustrate the relationship between laser pulse repetition rate and both shear strength and welding line width, respectively, at a constant laser power of 1.37W.

The shear strength exhibited an initial increase followed by a decrease with increasing repetition rate. As the repetition rate increased from 0.3 MHz to 1 MHz, the shear strength increased from 43.5 MPa to 147.6 MPa. However, at 1.2 MHz, shear strength sharply declined to 54.7 MPa. While a slight decrease was observed at 0.8 MHz compared to 0.5 MHz, the overall trend was an initial increase followed by a decrease. In contrast, welding line width decreased from 20 μm to 15 μm with increasing repetition rate, demonstrating a negative correlation. This behavior is attributable to the shorter pulse interval at higher repetition rates, where subsequent pulses interact with the pre-heated region from previous pulses. In high-repetition-rate welding, the molten pool experiences repeated melting and solidification cycles, with reduced cooling times and increased cooling rates, leading to a narrower weld line [37]. At a fixed laser output power, a decrease in the repetition rate increases the single-pulse energy. This can induce defects such as cracks and pores, thereby reducing the shear strength. Conversely, an increase in the repetition rate reduces the single-pulse energy, which mitigates defect formation. Simultaneously, the shorter pulse interval promotes more thorough material fusion, resulting in a more continuous, homogeneous, and dense weld zone with improved shear strength. However, an excessively high repetition rate leads to insufficient single-pulse energy (below the material modification threshold) and causes the pulse interval to become shorter than the material's thermal relaxation time. This results in excessive heat accumulation within the glass, generating considerable thermal stress that consequently reduces the shear strength.

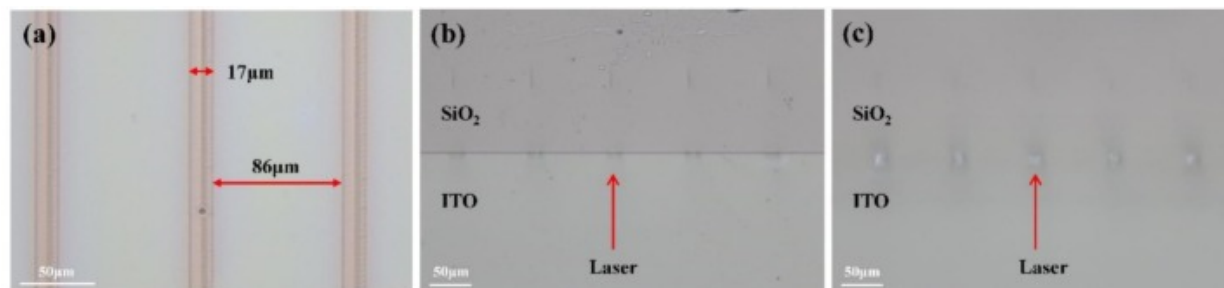
Fig. 5 (c) and (d) illustrate the relationship between laser pulse energy and both shear strength and weld line width, respectively, at a fixed repetition rate of 1 MHz. Shear strength initially increases and then decreases within the pulse energy range of 0.67 μJ to 2 μJ . Specifically, as the pulse energy increases from 0.67 μJ to 1.37 μJ , the shear strength reaches a peak value of 147.6 MPa. However, when the pulse energy is increased further to 2 μJ , the shear strength significantly drops to 41 MPa. Conversely, the weld line width consistently increases with increasing laser pulse energy, demonstrating a positive correlation between these two variables. This behavior is likely due to the higher energy

density per unit area at higher single-pulse energies, which expands the material melting zone and enhances heat conduction to the surrounding area, thus widening the weld line and initially improving shear strength within an optimal energy density range [38]. Beyond a pulse energy of 1.37 μJ , the weld seam width continues to expand, but the shear strength declines. This suggests that the effective welded area alone does not dictate shear strength. Laser ablation is another critical factor influencing weld quality. At low pulse energies, laser ablation is effectively suppressed within the irradiated region, resulting in minimal negative impact on shear strength. However, when the pulse energy exceeds the critical value of 1.37 μJ , laser ablation can no longer be effectively suppressed in the focal region [39]. Consequently, despite the increase in effective welded area, the shear strength decreases markedly beyond 1.37 μJ .

The shear strength and weld line width of joints between ITO glass and silica glass cover plates are depend on the pulse energy and repetition rate of the femtosecond laser. The experimental results indicate that a maximum shear strength of 147.6MPa was achieved with a pulse energy of 1.37 μJ and a repetition rate of 1 MHz.

Welding experiments were performed using a femtosecond laser (1.37 μJ , 1 MHz) to join perovskite-coated ITO glass substrates to silica glass cover plates. The interfacial morphology was systematically characterized using optical microscopy (OM) and scanning electron microscopy (SEM). Standard cross-sectional preparation using diamond-blade dicing and polishing can readily induce perovskite contamination and decomposition. Consequently, it fails to accurately represent the post-encapsulation structure, which is why our characterization focused on the quality of the welded interface. OM observations, as shown in Fig. 6(a), revealed well-defined weld lines on the ITO side, devoid of visible cracks between adjacent lines, indicating a high-quality joint. The weld line width was measured to be approximately 17 μm , with a spacing of approximately 86 μm between lines. Cross-sectional analysis of the welded sample, presented in Fig. 6 (b)-(c), shows a weld spacing of 10 μm in Fig. 6 (b), where the red arrow indicates the laser's incident direction. After femtosecond laser irradiation of the interface from the ITO glass side, clear molten traces were observed on the silica glass

side. Fig. 6 (c) demonstrates the effective elimination of the original interfacial gaps, with the formation of a continuous melting-recrystallization transition zone in the irradiated region [40]. This non-equilibrium phase transformation is considered strong evidence for covalent bond formation across the interface [29], further corroborating the effectiveness and reliability of femtosecond laser welding for joining glass-based materials.



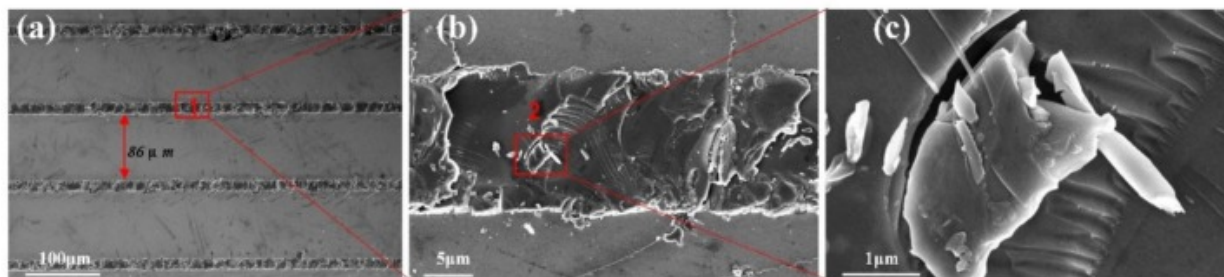
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Fig. 6. OM morphology of welded joints: (a) weld line surface, (b-c) weld line cross-sectional morphology.

To systematically evaluate the bonding quality of the welded interface, a mechanical shear force was applied to the welded silica glass and ITO glass sample to induce interface separation. Subsequently, the morphology of the separated ITO glass surface was examined in detail using SEM. Analysis of Fig. 7 (a) revealed an average weld width of 17 μm and a spacing of 86 μm. No evident spatter or ablation marks were observed in the non-irradiated region, and the uniform weld line width reflects high-precision laser scanning control and stable energy distribution. A magnified view of Region1 (Fig. 7 (b)) revealed significant melting and mixing of the two glass materials at the interface, confirming effective fusion. An enlargement of Region2 (Fig. 7 (c)) revealed micrometer-scale particulate structures associated with crack propagation, further demonstrating successful fusion between the ITO glass and silica glass at the microscopic level and indicating that interfacial fracture occurred under external force. This comprehensive

analysis demonstrates the feasibility of femtosecond laser welding for achieving high-quality glass joining.



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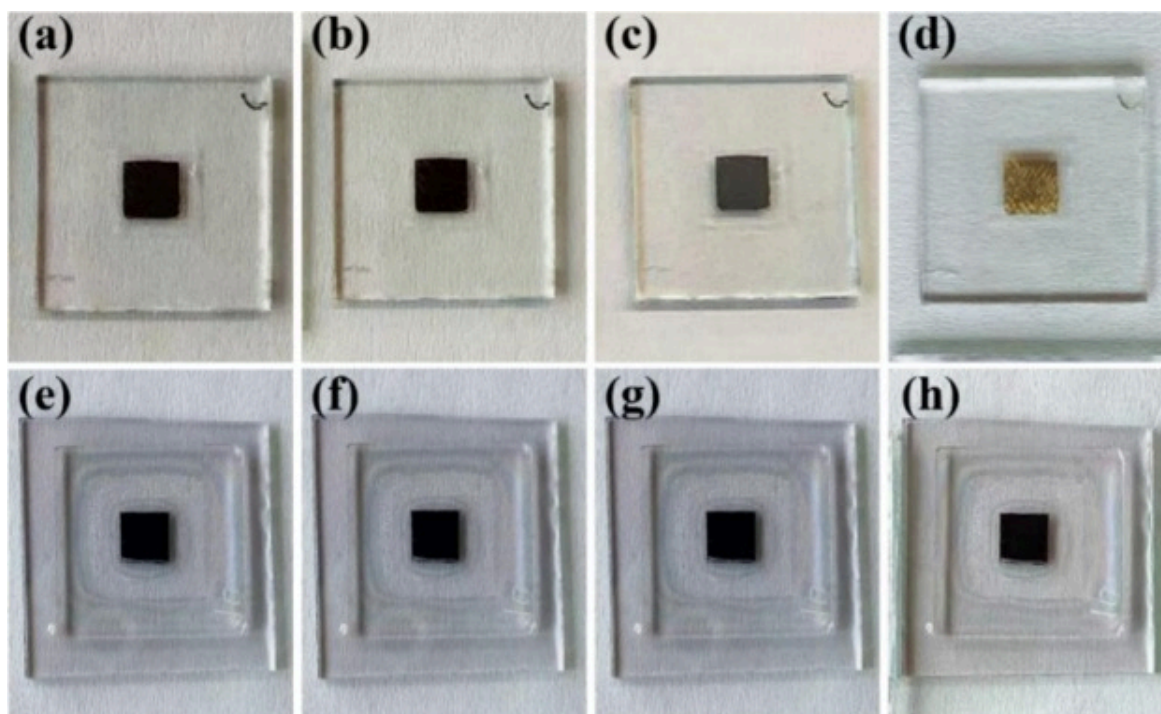
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Fig. 7. SEM of the sample surface after separation: (a) morphology of the ITO side, (b, c) higher-magnification views of areas 1 and 2.

The sealing performance of welded samples was evaluated by monitoring the colorimetric response of the encapsulated perovskite film, which changes from brown to yellow upon moisture exposure. Two methods were employed: first, sealed and unsealed samples were placed in a 60% relative humidity environment and subjected to thermal cycling (-20°C to 85°C , 30 min per cycle, 10 cycles). Second, the sealed samples were immersed in water (depth: 1 cm, temperature: 35°C , pressure: ambient) to directly assess and compare their waterproof sealing capability.

Both unencapsulated and encapsulated samples underwent thermal cycling tests, Fig. 8 (a)–(d) illustrate the color evolution of the perovskite film in the unencapsulated samples after 0, 3, 7, and 10 cycles, respectively, while Fig. 8 (e)–(h) depict the performance of the encapsulated samples under the same cycling conditions. As shown in Fig. 8 (c), obvious fading—manifested as significant lightening of the material color—was observed on the perovskite film surface of the unencapsulated sample after 7 thermal cycles, indicating the onset of film degradation. By 10 cycles, the film had developed pronounced yellowing (Fig. 8 (d)), a typical visual indicator of the hydrolysis reaction in perovskite materials,

confirming structural damage due to moisture intrusion from the environment. In the encapsulated sample, Newton's rings were clearly visible along the periphery of the sealing line (Fig. 8 (e)). Their presence provides direct evidence of a sub-micron parallel gap at the sealed interface, indicating that the femtosecond laser welding process achieved sealing interfaces with nanoscale flatness and excellent hermetic bonding at the silica/ITO glass interface. Crucially, analysis of Fig. 8 (e)–(h) reveals that no yellowing or noticeable color change occurred in the inner perovskite film of the encapsulated sample, even after 10 thermal cycles. These findings confirm that the laser-welded encapsulation layer effectively blocks environmental moisture penetration, thereby significantly enhancing the environmental stability of the perovskite film.



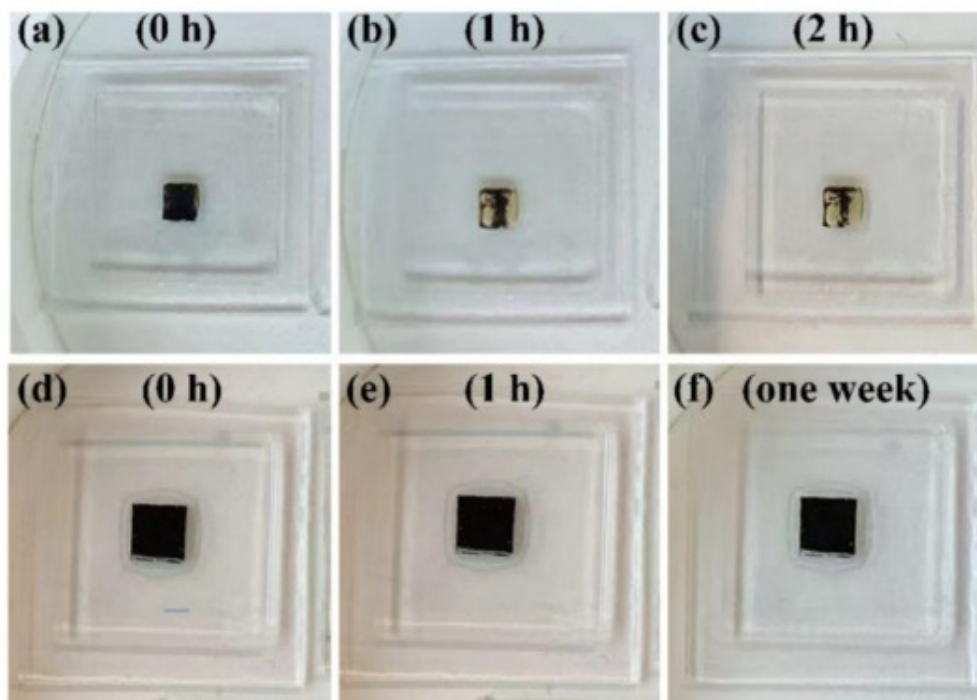
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Fig. 8. Comparison of perovskite material stability under thermal cycling: (a-d) unsealed samples after 0, 3, 7, and 10 cycles; (e-h) silica-glass-sealed samples after 0, 3, 7, and 10

cycles.

Water immersion tests on fully sealed and incompletely sealed samples yielded the following results. As shown in Fig. 9 (a), the perovskite film at the edge region of the incompletely sealed sample began to discolor shortly after immersion. Over time, the discolored area expanded from the edge toward the center, eventually causing the entire film to turn yellow, as shown in Fig. 9 (b)–(c). This indicates that moisture rapidly penetrated the sample through interfacial defects, resulting in extensive degradation of the perovskite layer. In contrast, Fig. 9 (d)–(f) present the experimental results of the fully sealed sample under the same conditions. After 168 h of continuous immersion, the perovskite film exhibited no significant color change, and the edges remained intact without any discoloration (Fig. 9 (f)). These results demonstrate that the laser-sealed interface effectively blocks the permeation of water and oxygen, significantly enhancing the environmental stability and durability of the perovskite film.



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Fig. 9. Water resistance test of a perovskite thin film covered with silica glass.(a)-(c)show the sealing effect of an incomplete encapsulation, while(d)-(f)show the effect of a complete encapsulation.

The results demonstrate that the perovskite film in the hermetically sealed sample showed no detectable color change after undergoing 10 thermal cycles and 168h of water immersion. This highlights the exceptional performance stability of samples sealed using the femtosecond laser technique. These results establish a robust experimental foundation and technical approach for further improving the sealing process and enhancing the long-term stability and reliability of perovskite film-based devices.

4. Conclusions

This study presents demonstration of encapsulating perovskite films by direct femtosecond laser (1030nm wavelength, 300fs pulse width) welding of silica glass. By controlling the laser parameters (single pulse energy of 1.37 μ J, repetition rate of 1 MHz), a micron-scale melting-recrystallization process was induced at the interface between the indium tin oxide (ITO) glass and the silica glass. This process resulted in high-strength hermetic seals with a shear strength of up to 147.6MPa. This effectiveness of this encapsulation method was confirmed: the encapsulated perovskite films showed no signs of degradation after 10 cycles of thermal cycling (between 85°C and -20°C) and 168h of water immersion. This technique effectively mitigates the failure mechanisms associated with aging-induced volatilization in conventional resin-based encapsulation and surpasses the stability limitations of picosecond laser sealing.

We have successfully achieved effective electrical conduction in the encapsulated cells. The long-term stability evaluation and systematic performance analysis of these cells will be detailed in subsequent work. This technology provides a practical solution for the

long-term stable operation of environmentally sensitive optoelectronic devices, particularly PSCs.

CRediT authorship contribution statement

Liangjing Wang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Shuang Liu:** Writing – review & editing, Supervision, Resources. **Huan Zhan:** Resources, Project administration. **Guanghua Cheng:** Writing – review & editing, Resources, Methodology. **Ru Li:** Writing – review & editing, Supervision, Resources. **Jian Liang:** Project administration. **Yingwei Liu:** Project administration. **Haisong Tang:** Visualization, Software. **Yuhao Wu:** Writing – review & editing, Formal analysis, Data curation. **Li Lu:** Writing – review & editing, Supervision. **Jinjin Yang:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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